

Yamini Karthik

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EDUCATION

San Jose State University - B.S. in Computer Science | Expected Dec 2027

- o GPA: 3.89/4.00, [President's Scholar](#), [Dean's Scholar](#)
- o **Related Coursework:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Data Visualization, Formal Languages & Computability, Discrete Mathematics, Calculus I & II, Linear Algebra

EXPERIENCE

Cadence Design Systems - Software Engineering Sprintern

Jul 2025 - Aug 2025

- Designed & implemented a scalable **HTML-to-PDF pipeline** in **Python** using **Playwright** & **PyMuPDF**, processing **100+ HTML files** and automating technical document generation.
- Automated PDF generation with dynamic **TOCs**, **hierarchical bookmarks**, & **hyperlink preservation**, reducing production time by **80%**.
- Conducted **performance testing** and debugging to address memory limitations and document navigation issues, validating the pipeline for production use; presented technical implementation and optimization results to Cadence engineers.

Latitude AI - AI Studio Fellow

Aug 2025 - Dec 2025

- Benchmarked **Mask R-CNN (ResNet-50 FPN)** against **Segment Anything Model (SAM)** using **100K+ BDD100K/Cityscapes images** for autonomous driving perception.
- Built **data processing and evaluation workflows** comparing **Mask R-CNN** and foundation models, identifying trade-offs in segmentation quality and deployment performance.
- Presented performance insights and deployment considerations to industry stakeholders.

Break Through Tech - AI/ML Fellow

May 2025 - Apr 2026

- Selected from **3,500+ applicants** for a competitive 12-month AI fellowship.
- Completed Cornell Machine Learning Foundations coursework ([Machine Learning Foundations Course](#)) as well as applied machine learning projects through AI Studio and the Women in Data Science (WiDS) Datathon.

PROJECTS

[AI Travel Itinerary Planner](#)

Aug 2026 - Sep 2026

DevNetwork {API + Cloud + AI} Hackathon 2026 - Hackathon Project

- Architected a full-stack **AI travel planning application** using **React + FastAPI**, integrating **Gemini** and **SerpApi** to validate user preferences and retrieve real-time flights, hotels, attractions, and restaurants.
- Engineered **preference-aware retrieval and structured itinerary generation**, combining external travel data with LLM-generated JSON to produce budget-, schedule-, dietary-, and trip-type-aware daily itineraries.
- Implemented **Pydantic validation**, **REST APIs**, **frontend routing**, and **localStorage persistence**, separating third-party API access behind the backend and enabling users to save, search, and revisit itineraries.

[Agentic Skincare & Haircare Advisor](#)

May 2026 - Aug 2026

Agentic AI Innovation Challenge 2026 - Hackathon Submission

- Architected and deployed a full-stack AI application using **React**, **FastAPI**, and a **Gemini tool-calling agent**, implementing REST APIs for conversational interactions, profile management, ingredient analysis, and personalized recommendations.
- Engineered a **modular backend** separating API routing, agent orchestration, business logic, and data processing, with structured profiles, provenance tracking, and backend-owned session state.
- Integrated deterministic tools for **product retrieval**, ingredient compatibility analysis, and recommendation generation, separating business logic from LLM reasoning to improve maintainability and transparency.

ACTIVITIES AND LEADERSHIP

Responsible Computing Club - Mozilla Workshop Ambassador/Mozilla Industry Ambassador

Sep 2024 - May 2026

- Organized networking events and professional speaker panels by engaging industry experts.
- Partnered with club leadership to develop **10+ workshops** on AI ethics, facial recognition, and emerging technologies.

SKILLS

Programming: Java, Python, JavaScript, HTML/CSS, C, SQL

Frameworks: React, FastAPI, PyTorch, Scikit-Learn, Pandas, NumPy,

Tools: Git, SQLite, Jupyter Notebooks, Matplotlib, IntelliJ, VS Code, Figma